

PROJECT REPORT

Optimizing Loan Eligibility ~ A study of Machine Learning Models

Team – Chandrika Kunchakuri & Satya Sai Gayathri Palacherla

Background:

Loan services play a crucial role in the modern world, serving as a primary source of profit for financial institutions. They help students cover educational and living expenses and enable individuals to acquire luxuries like homes and cars. However, determining whether an applicant qualifies for a loan requires careful evaluation of various factors. The "Loan Prediction Problem Dataset," sourced from Kaggle by Debdatta Chatterjee, offers a comprehensive set of metrics to analyze the determinants of loan approval. This study focuses on identifying the most effective metric for loan approval using logistic regression and random forest models.

Dataset Attributes:

The data collection consists of 614 records and 13 attributes, which include:

Loan_ID: A unique identification number for each loan application.

Gender: The gender of the applicant (Male or Female).

Married: The marital status of the applicant (Yes or No).

Dependents: The number of dependents of the applicant (0, 1, 2, or 3+).

Education: The level of education of the applicant (Graduate or Not Graduate).

Self_Employed: Whether the applicant is self-employed or not (Yes or No).

ApplicantIncome: The monthly income of the applicant.

CoapplicantIncome: The monthly income of the co-applicant.

LoanAmount: The loan amount requested (in thousands).

Loan_Amount_Term: The term of the loan, in months.

Credit_History: A binary variable representing the applicant's credit history (1 if they have a credit history that meets the guidelines, 0 otherwise).

Property_Area: The category of the property area where the applicant resides (Urban, Semiurban, or Rural).

Loan_Status: The final decision on the loan application (Y for approved, N for not approved).

Data Cleaning and Preparation:

Our Data:

```
'data.frame': 614 obs. of 13 variables:
 $ Loan_ID      : Factor w/ 614 levels "LP001002","LP001003",...: 1 2 3 4 5 6 7 8 9
10 ...
 $ Gender       : Factor w/ 3 levels "", "Female", "Male": 3 3 3 3 3 3 3 3 3 ...
 $ Married      : Factor w/ 3 levels "", "No", "Yes": 2 3 3 3 2 3 3 3 3 ...
 $ Dependents   : Factor w/ 5 levels "", "0", "1", "2",...: 2 3 2 2 2 4 2 5 4 3 ...
 $ Education    : Factor w/ 2 levels "Graduate", "Not Graduate": 1 1 1 2 1 1 2 1 1 1
...
 $ Self_Employed : Factor w/ 3 levels "", "No", "Yes": 2 2 3 2 2 3 2 2 2 ...
 $ ApplicantIncome : int  5849 4583 3000 2583 6000 5417 2333 3036 4006 12841 ...
 $ CoapplicantIncome: num  0 1508 0 2358 0 ...
 $ LoanAmount     : int  NA 128 66 120 141 267 95 158 168 349 ...
 $ Loan_Amount_Term : int  360 360 360 360 360 360 360 360 360 360 ...
 $ Credit_History  : int   1 1 1 1 1 1 1 0 1 1 ...
 $ Property_Area   : Factor w/ 3 levels "Rural", "Semiurban",...: 3 1 3 3 3 3 3 2 3 2
...
 $ Loan_Status    : Factor w/ 2 levels "N", "Y": 2 1 2 2 2 2 2 1 2 1 ...
```

During the data cleaning process, we identified several columns with missing values and decided to drop them. In the "Dependents" column, the value "3+" was replaced with "3" for consistency. Additional preprocessing steps included converting specific columns into categorical factors to better represent their nature. The "Loan_Amount_Term" column was removed, as it was largely uniform (predominantly 360) and deemed irrelevant for analysis.

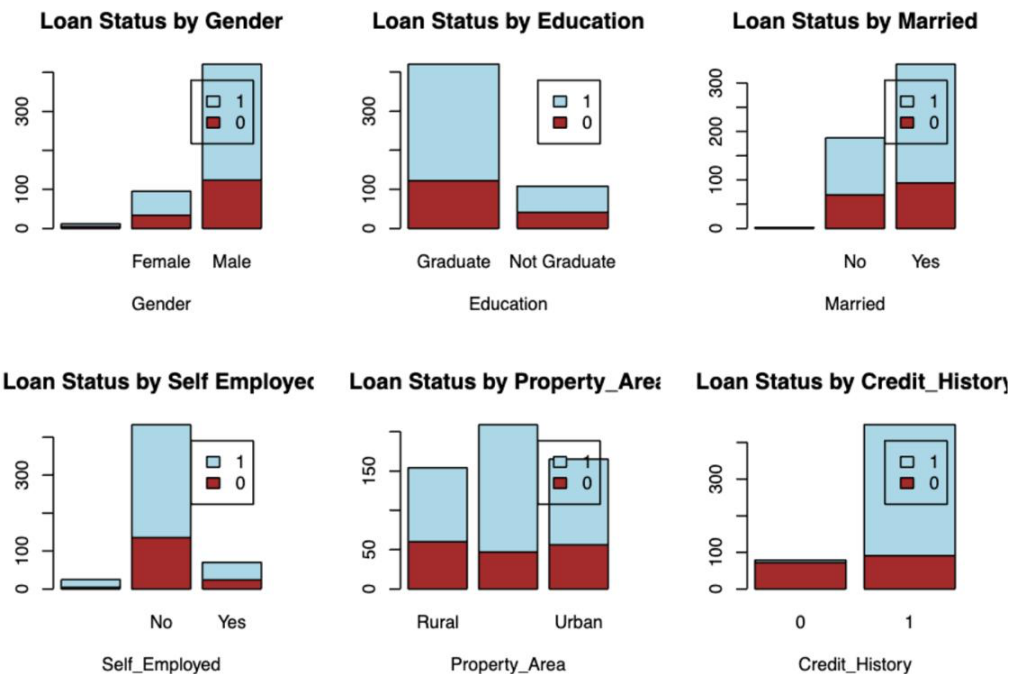
We further created a binary indicator variable for loan approval ("yes"), removed the original loan status column, and transformed the "ApplicantIncome" column into numeric format to facilitate further analysis. After these adjustments, the dataset was reduced to 528 rows.

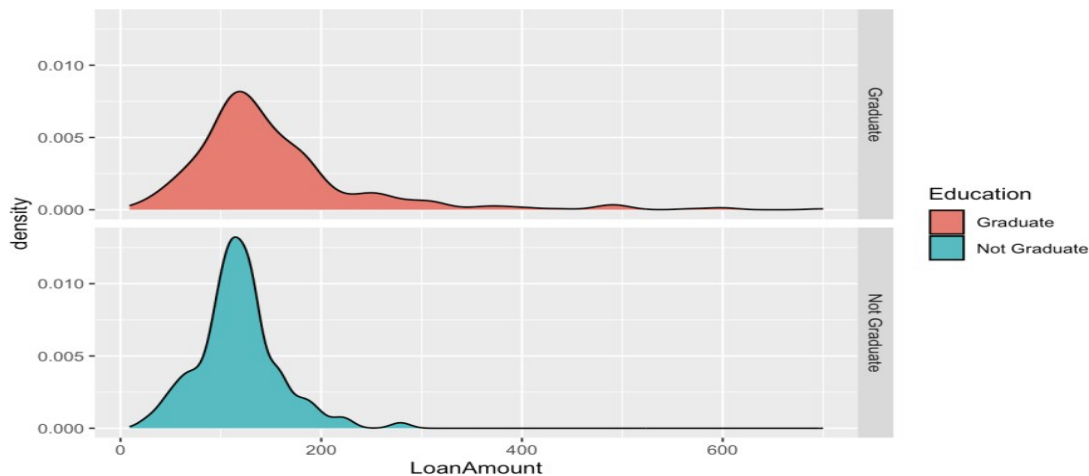
Overview of the prepared data:

```
'data.frame': 528 obs. of 11 variables:
 $ Gender      : Factor w/ 3 levels "", "Female", "Male": 3 3 3 3 3 3 3 3 3 3 ...
 $ Married     : Factor w/ 3 levels "", "No", "Yes": 3 3 3 2 3 3 3 3 3 3 ...
 $ Dependents  : num 3 2 2 2 4 2 3 4 3 4 ...
 $ Education   : Factor w/ 2 levels "Graduate", "Not Graduate": 1 1 2 1 1 2 1 1 1 1 ...
 $ Self_Employed : Factor w/ 3 levels "", "No", "Yes": 2 3 2 2 3 2 2 2 2 2 ...
 $ ApplicantIncome : num 4583 3000 2583 6000 5417 ...
 $ CoapplicantIncome: num 1508 0 2358 0 4196 ...
 $ LoanAmount   : num 128 66 120 141 267 95 158 168 349 70 ...
 $ Credit_History : Factor w/ 2 levels "0", "1": 2 2 2 2 2 2 1 2 2 2 ...
 $ Property_Area : Factor w/ 3 levels "Rural", "Semiurban", "...: 1 3 3 3 3 3 2 3 2 3 ...
 $ yes         : Factor w/ 2 levels "0", "1": 1 2 2 2 2 2 1 2 1 2 ...
```

Insights from Exploratory Data Analysis:

Here is the comparison plot between loan status and all other variables





Upon examining the Gender graph, we observe that males constitute a larger proportion of the data collection, with more than half of their loan applications being approved. Although there are fewer female applicants, a significant portion of their applications have also been approved. We look at the other charts with the same eye to evaluate how each category performed in regards to the approval of the loan applications.

Choosing Predictive Models:

Training and Testing Split:

The data collection is divided into two parts: 70% as train data and the remaining 30% as test data. We'll build the approaches on the train data and text its error on the test data.

Logistic Regression Analysis:

We used all the predictors in this approach.

Summary: |

Train	0	1
0	50	65
1	10	244

Sensitivity for Train Data	Specificity for Train Data
0.9606299	0.4347826

Test	0	1
0	2	0
1	1	8

Sensitivity for Test Data	Specificity for Test Data
0.8888889	1

```
Call:
glm(formula = yes ~ ., family = binomial, data = train_data)

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)  1.191e+01  6.136e+02   0.019  0.984512
GenderFemale -1.341e+00  1.177e+00  -1.139  0.254724
GenderMale   -6.623e-01  1.111e+00  -0.596  0.550917
MarriedNo    -1.311e+01  6.136e+02  -0.021  0.982950
MarriedYes   -1.270e+01  6.136e+02  -0.021  0.983482
Dependents   -2.713e-04  1.965e-01  -0.001  0.998898
EducationNot Graduate -5.904e-01  3.474e-01  -1.700  0.089203
Self_EmployedNo -6.133e-01  6.997e-01  -0.876  0.380780
Self_EmployedYes -1.200e+00  7.797e-01  -1.539  0.123741
ApplicantIncome 2.458e-05  2.934e-05   0.838  0.402224
CoapplicantIncome 1.746e-05  6.880e-05   0.254  0.799634
LoanAmount     -4.385e-03  2.347e-03  -1.868  0.061722
Credit_History1 3.678e+00  5.260e-01   6.994  2.67e-12 ***
Property_AreaSemiurban 1.316e+00  3.569e-01   3.688  0.000226 ***
Property_AreaUrban 4.701e-01  3.293e-01   1.428  0.153382
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 457.87  on 368  degrees of freedom
Residual deviance: 336.43  on 354  degrees of freedom
AIC: 366.43

Number of Fisher Scoring iterations: 13
```

With a accuracy in identifying positive cases of approximately 96.06 %, the approach effectively captures the vast majority of actual positive cases in the training set.

While the accuracy in identifying negative cases is still 43.48% the he approach's ability to detect negative instances is comparatively weaker in the training data, where it correctly identifies fewer negative cases, Conversely, the accuracy in identifying negative cases for the test data is 100%, implying that the approach accurately identifies all negative instances in the test set

Odds ratio: (Acc to the importance factor – The variables were choosen)

```
Credit_History1
39.58476
ApplicantIncome
1.000025
CoapplicantIncome
1.000017
LoanAmount
0.9956251
Property_AreaSemiurban
3.729264
```

Credit_History: Reflects an applicant’s credit history. Good credit history is favorable for loan approval, while insufficient credit history may pose challenges.

ApplicantIncome: Represents the primary applicant’s income. Higher income improves loan approval chances.

CoapplicantIncome: Includes earnings from other contributors (e.g., spouse). Combined income impacts loan eligibility.

LoanAmount: Indicates the requested loan value. Lenders assess it relative to income.

Property_Area (Semiurban): Describes the property location. Semiurban areas have distinct risk profiles for loan approval.

Backward Selection:

Let's consider backward stepwise selection, to find a approach with much more significance. The approach is: $\text{yes} \sim \text{Gender} + \text{Education} + \text{Loan_Amount} + \text{Credit_History} + \text{Property_Area}$.

Train	0	1
0	50	65
1	7	247

Sensitivity for Train Data	Specificity for Train Data
0.9724409	0.4347826

Test	0	1
0	2	0
1	1	8

Sensitivity for Test Data	Specificity for Test Data
0.8888889	1

Summary:

$\text{yes} \sim \text{Gender} + \text{Education} + \text{LoanAmount} + \text{Credit_History} + \text{Property_Area}$

Call:

```
glm(formula = yes ~ Gender + Education + LoanAmount + Credit_History +  
    Property_Area, family = binomial, data = train_data)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-1.558723	1.322782	-1.178	0.238650
GenderFemale	-1.562279	1.174315	-1.330	0.183395
GenderMale	-0.718615	1.128102	-0.637	0.524117
EducationNot Graduate	-0.591952	0.342829	-1.727	0.084228 .
LoanAmount	-0.002877	0.001681	-1.711	0.087070 .
Credit_History1	3.624852	0.513431	7.060	1.66e-12 ***
Property_AreaSemiurban	1.362252	0.351300	3.878	0.000105 ***
Property_AreaUrban	0.524563	0.320722	1.636	0.101930

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 457.87 on 368 degrees of freedom
Residual deviance: 342.75 on 361 degrees of freedom
AIC: 358.75

Number of Fisher Scoring iterations: 5

The summary indicates that most insignificant predictors have been excluded. The approach demonstrates a strong ability to identify positive cases, with an accuracy of approximately

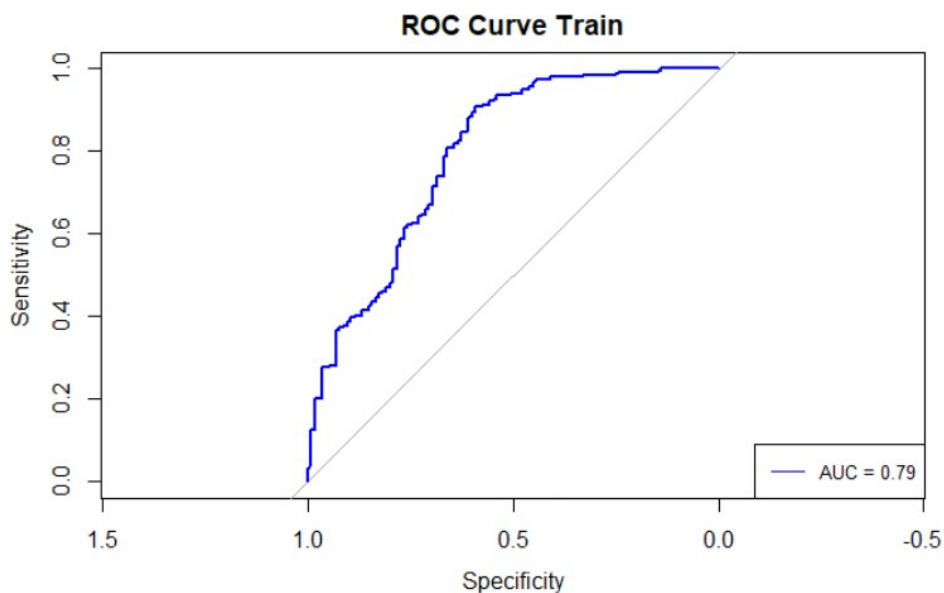
97.24% in the training set. However, its accuracy in detecting negative cases is lower at 43.48%, showing weaker performance in correctly identifying negative instances within the training data. In contrast, the test data shows 100% accuracy for negative cases, indicating that the approach effectively identifies all negative instances in this subset..

Odds Ratio:

Credit_History1	37.51919
LoanAmount	0.9971272
Property_AreaSemiurban	3.904978

A good credit history significantly improves loan approval chances. Loan amount has a minimal impact on approval odds. Semi-urban property areas are favorable for loan approval.

Roc Curve:



The AUC value of approximately 0.79 suggests that the logistic regression approach has good discriminatory power in distinguishing between approved and rejected loan applications on the training data collection.

Random Forest Analysis:

Random Forest is an ensemble machine learning technique that can be applied to both classification and regression problems. It works by generating multiple decision trees during the training phase and combines their outputs, using the majority class for classification tasks or the average prediction for regression tasks.

In random forest we implemented 3 approaches and using the tree count 500 which are given by:

Model 1:

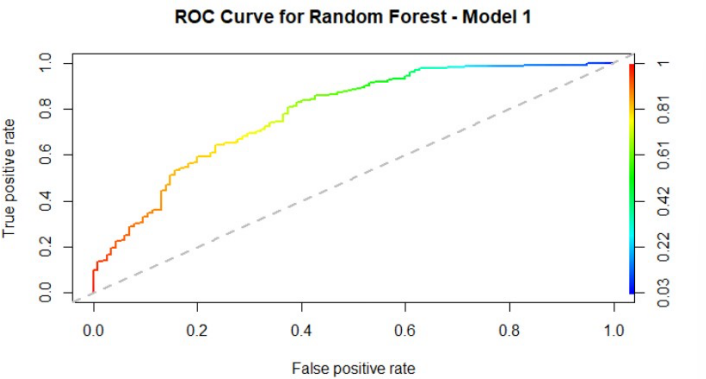
```
rf_approach1 <- randomForest(myyes_train ~ ., data = train_data[,-11])
```

Train	0	1
0	113	2
1	0	254
Sensitivity for Train Data		Specificity for Train Data
1		0.9826087

Test	0	1
0	2	0
1	2	7
Sensitivity for Test Data		Specificity for Test Data
0.7777778		1

In conclusion, the approach shows excellent performance on both the training and test datasets, with high accuracy in detecting positive and negative cases. However, a slight reduction in accuracy for positive cases is noted in the test dataset compared to the training dataset, indicating the approach may generalize slightly less effectively to unseen data. Overall, the performance is strong, with an AUC value of 0.7791.

ROC Curve for approach 1:



Model 2:

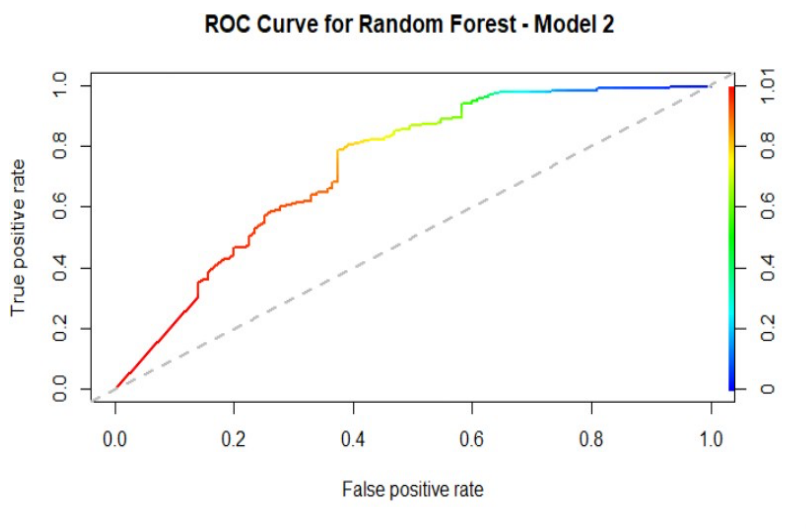
yes~ Gender+ Education + Loan_Amount + Credit_History + Property_Area.

Train	0	1
0	58	57
1	5	249
Sensitivity for Train Data		Specificity for Train Data
0.980315		0.5043478

Test	0	1
0	2	0
1	1	8
Sensitivity for Test Data		Specificity for Test Data
0.8888889		1

Model 2 shares the same parameters with those selected through backward selection in logistic approach.

In summary, the approach exhibits strong performance in accurately detecting positive cases in both the training and test datasets. However, it struggles with lower accuracy in identifying negative cases in the training data, reflecting some challenges in correctly classifying these instances. The AUC value of 0.7406 indicates a moderate overall ability to distinguish between positive and negative cases.



Model Comparison:

Both the approaches perform a little similar to each other on training set and exactly the same way on testing data.

Model 3:

myyes_train ~ ApplicantIncome + CoapplicantIncome + LoanAmount + Credit_History + Property_Area Model 3 was built on the factor called importance table for the Model 1 i.e. importance(rf_approach1)

	MeanDecreaseGini
Gender	5.012530
Married	4.307860
Dependents	7.802977
Education	4.127145
Self_Employed	4.648639
ApplicantIncome	30.789496
CoapplicantIncome	16.763052
LoanAmount	28.189683
Credit_History	35.067484
Property_Area	9.874131

Train	0	1
0	97	18
1	0	254

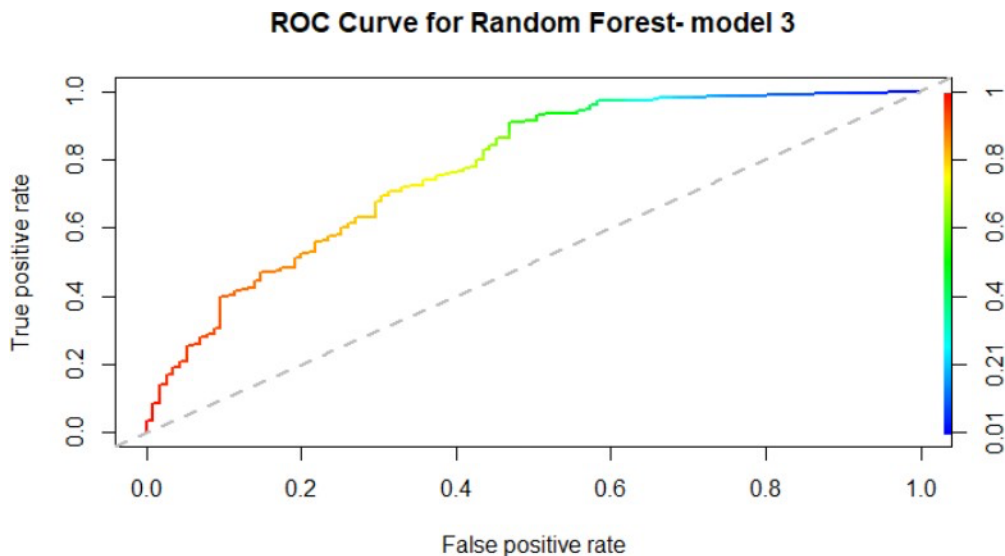
Sensitivity for Train Data	Specificity for Train Data
1	0.8434783

Test	0	1
0	2	0
1	0	9

Sensitivity for Test Data	Specificity for Test Data
1	1

The top 5 important factors from the table were considered in-order to build the approach:

The approach shows exceptional performance on both the training and test datasets. It achieves perfect accuracy in identifying positive cases across both sets, correctly classifying all positive instances. Furthermore, it demonstrates high accuracy in detecting negative cases, accurately identifying a substantial portion of them. The flawless results on the test set, coupled with a moderately high AUC of 0.7718, underscore the approach's robustness and effectiveness in binary classification tasks.



Lasso Regression:

Summary:

```
Best lambda (regularization parameter): 0.01406281
Regression Coefficients at Best Lambda:
19 x 1 sparse Matrix of class "dgCMatrix"
      s0
(Intercept) -2.103731e+00
GenderFemale .
GenderMale .
MarriedNo -2.212567e-01
MarriedYes .
Dependents0 .
Dependents1 -2.675514e-01
Dependents2 .
Dependents3+ .
EducationNot Graduate .
Self_EmployedNo .
Self_EmployedYes .
ApplicantIncome .
CoapplicantIncome -3.097464e-05
LoanAmount -2.425505e-04
Loan_Amount_Term .
Credit_History 3.460861e+00
Property_AreaSemiurban 3.876392e-01
Property_AreaUrban .
Selected Features (non-zero coefficients):
[1] "(Intercept)" "MarriedNo" "Dependents1" "CoapplicantIncome" "LoanAmount"
[6] "Credit_History" "Property_AreaSemiurban"
```

```
Confusion Matrix:
Confusion Matrix and Statistics

      Reference
Prediction 0 1
0 16 2
1 20 84

      Accuracy : 0.8197
      95% CI : (0.7398, 0.8834)
      No Information Rate : 0.7049
      P-Value [Acc > NIR] : 0.0026493

      Kappa : 0.4928

      Mcnemar's Test P-Value : 0.0002896

      Sensitivity : 0.4444
      Specificity : 0.9767
      Pos Pred Value : 0.8889
      Neg Pred Value : 0.8077
      Prevalence : 0.2951
      Detection Rate : 0.1311
      Detection Prevalence : 0.1475
      Balanced Accuracy : 0.7106

      'Positive' Class : 0

Model Accuracy: 0.8196721
```

1. Key Predictors of Loan Approval:

- Credit History: The most influential factor; applicants with a positive credit history are significantly more likely to have their loans approved.
- Property Area: Living in semi-urban areas positively affects loan approval.
- Loan Amount: Higher loan amounts reduce approval chances, though the impact is small.
- Dependents: Having dependents slightly decreases approval likelihood.

2. Model Performance:

- Accuracy: The model correctly predicts loan outcomes 81.97% of the time.
- Specificity (Approved Loans): High at 97.67%, indicating excellent detection of approved loans.
- Sensitivity (Rejected Loans): Low at 44.44%, suggesting difficulty in identifying rejected loans.
- AUC: The model achieves a fair discriminatory power with an AUC of ~ 0.71 .

3. Insights:

- Credit history is the most critical feature for loan decisions.
- Semi-urban property areas and being unmarried increase approval chances slightly.
- The model underperforms in predicting rejected loans, requiring improvement in this area.

Summary of Findings:

This project focused on predicting loan approvals using three machine learning approaches: logistic regression, random forest, and Lasso regression. The goal was to identify key factors influencing loan decisions and assess the performance of each model.

Key findings:

1. Important Predictors:

- Credit History was identified as the most influential factor for loan approval.
- Property Area (specifically semi-urban) had a positive impact on approval chances.
- Loan Amount and Dependents negatively affected approval, though their impact was relatively small.

2. Model Performance:

- Logistic Regression showed strong accuracy for predicting approved loans but struggled with rejected loans. Refining the model with backward selection improved its ability to identify rejected loans.
- Random Forest demonstrated robust performance, with the third model achieving flawless results by focusing on the most important features.

- Lasso Regression efficiently identified significant predictors, achieving good accuracy (81.97%) but facing challenges in predicting rejected loans due to low sensitivity.

3. Insights and Challenges:

- Feature selection played a critical role in improving model performance, particularly with credit history and property area.
- While logistic regression offered interpretability and random forest provided robustness, Lasso regression optimized feature selection for better model simplicity.
- The models faced challenges in predicting rejected loans, pointing to the need for addressing class imbalance.

Conclusion:

The project demonstrated the importance of using machine learning for loan prediction, showing that credit history and property area are critical for determining loan approval. Although the models performed well in identifying approved loans, improving sensitivity for rejected loans remains an area for enhancement. The findings emphasize the need for further model refinement and the integration of techniques like oversampling to handle class imbalances.

Literature Review:

Reference 1: Credit Risk Assessment and Financial Decision Support Using Explainable Artificial Intelligence” published by MDPI. This research discusses the application of machine learning models, including logistic regression and random forest, for credit risk assessment, emphasizing the importance of model interpretability in financial decision-making. The study uses Shapley values for explainability, enhancing model transparency in credit risk management. We can find more information in the URL "<https://www.mdpi.com/2227-9091/12/10/164>".

Reference 2: A Comparison of Random Forest and Logistic Regression Model in Credit Scoring of Rural Households.” This study compares the effectiveness of random forest and logistic regression models in predicting creditworthiness for rural households, focusing on the interpretability of logistic regression and the predictive accuracy of random forest. For more information, we can visit the URL "https://www.researchgate.net/profile/Do-Nhung-7/publication/353326832_A_COMPARISON_OF_RANDOM_FOREST_AND_LOGISTIC_REGRESSION_MODEL_IN_CREDIT_SCORING_OF_RURAL_HOUSEHOLDS/links/60f47053f6b568a7098bd11e2/A-COMPARISON-OF-RANDOM-FOREST-AND-LOGISTIC-REGRESSION-MODEL-IN-CREDIT-SCORING-OF-RURAL-HOUSEHOLDS.pdf?origin=publication_detail".